WORKED SOLUTION

STEP II 2016, Question 7

Pure • Definite Integrals by Symmetry

STEP II 2016, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q7

The key idea

Reflecting the interval $[0, a]$ about its midpoint, $x \mapsto a - x$, doesn't change the integral, but it can change the integrand into something that combines beautifully with the original. For trigonometric integrands on $[0, \frac{1}{2}\pi]$ this swaps sin and cos; on $[0, \frac{1}{4}\pi]$ the compound-angle formula for $\tan(\frac{1}{4}\pi - x)$ does the work. The final part looks new, but an integration by parts turns it into part (iii).

Solution

The result $(*)$

In $\int_0^a f(a-x)\,\mathrm{d}x$ substitute $u = a - x$, so $\mathrm{d}u = -\mathrm{d}x$; when $x = 0$, $u = a$, and when $x = a$, $u = 0$. Then

$$\int_0^a f(a-x)\,\mathrm{d}x = \int_a^0 f(u)\,(-\mathrm{d}u) = \int_0^a f(u)\,\mathrm{d}u = \int_0^a f(x)\,\mathrm{d}x,$$

since the name of the integration variable doesn't matter. This is $(*)$.

Part (i)

Call the integral I. With $a = \frac{1}{2}\pi$ we have $\sin(\frac{1}{2}\pi - x) = \cos x$ and $\cos(\frac{1}{2}\pi - x) = \sin x$, so $(*)$ gives

$$I = \int_0^{\frac{1}{2}\pi} \frac{\cos x}{\sin x + \cos x}\,\mathrm{d}x.$$

Adding this to the original form of I,

$$2I = \int_0^{\frac{1}{2}\pi} \frac{\sin x + \cos x}{\cos x + \sin x}\,\mathrm{d}x = \int_0^{\frac{1}{2}\pi} 1\,\mathrm{d}x = \frac{\pi}{2},$$

so

$$I = \frac{\pi}{4}.$$

(The denominator $\cos x + \sin x$ is positive on $[0, \frac{1}{2}\pi]$, so nothing goes wrong.)

Part (ii)

Now $a = \frac{1}{4}\pi$, and the trick of part (i) no longer swaps sin and cos. Instead, dividing top and bottom by $\cos x$ (positive on this interval), the integrand is

$$\frac{\sin x}{\cos x + \sin x} = \frac{\tan x}{1 + \tan x}.$$

Let $t = \tan x$. Since $\tan(\frac{1}{4}\pi - x) = \dfrac{1-t}{1+t}$,

$$\frac{\tan(\frac{1}{4}\pi - x)}{1 + \tan(\frac{1}{4}\pi - x)} = \frac{(1-t)/(1+t)}{2/(1+t)} = \frac{1-t}{2} = \frac{1}{2} - \frac{1}{2}\tan x.$$

So by $(*)$, calling the integral J,

$$\begin{aligned} J &= \int_0^{\frac{1}{4}\pi} \Big(\frac{1}{2} - \frac{1}{2}\tan x\Big)\,\mathrm{d}x = \frac{\pi}{8} - \frac{1}{2}\Big[-\ln\cos x\Big]_0^{\frac{1}{4}\pi} \\ &= \frac{\pi}{8} + \frac{1}{2}\ln\frac{1}{\sqrt{2}} = \frac{\pi}{8} - \frac{1}{4}\ln 2. \end{aligned}$$

Ask yourself Is there a way to do part (ii) that doesn't use $(*)$ at all?

Yes. Let $K = \int_0^{\frac{1}{4}\pi} \frac{\cos x}{\cos x + \sin x}\,\mathrm{d}x$. Then $J + K = \frac{1}{4}\pi$, and since the numerator $\cos x - \sin x$ is the derivative of the denominator,

$$K - J = \Big[\ln(\cos x + \sin x)\Big]_0^{\frac{1}{4}\pi} = \ln\sqrt{2} = \tfrac{1}{2}\ln 2.$$

Subtracting, $2J = \frac{1}{4}\pi - \frac{1}{2}\ln 2$, which agrees with the answer above.

Part (iii)

Call the integral L. The same compound-angle formula gives

$$1 + \tan(\tfrac{1}{4}\pi - x) = 1 + \frac{1 - \tan x}{1 + \tan x} = \frac{2}{1 + \tan x}.$$

So by $(*)$,

$$L = \int_0^{\frac{1}{4}\pi} \ln\frac{2}{1+\tan x}\,\mathrm{d}x = \int_0^{\frac{1}{4}\pi} (\ln 2 - \ln(1+\tan x))\,\mathrm{d}x = \frac{\pi}{4}\ln 2 - L.$$

Hence $2L = \frac{1}{4}\pi\ln 2$ and

$$L = \frac{\pi}{8}\ln 2.$$

Part (iv)

Divide top and bottom by $\cos^2 x$:

$$\frac{x}{\cos x\,(\cos x + \sin x)} = \frac{x\sec^2 x}{1+\tan x}.$$

Ask yourself What has $\dfrac{\sec^2 x}{1+\tan x}$ got to do with part (iii)?

It is exactly the derivative of $\ln(1+\tan x)$, because $\frac{\mathrm{d}}{\mathrm{d}x}(1+\tan x) = \sec^2 x$; and $1+\tan x > 0$ on $[0, \frac{1}{4}\pi]$. So we integrate by parts with $u = x$ and $\mathrm{d}v = \dfrac{\sec^2 x}{1+\tan x}\,\mathrm{d}x$, $v = \ln(1+\tan x)$:

$$\begin{aligned}\int_0^{\frac{1}{4}\pi} \frac{x\sec^2 x}{1+\tan x}\,\mathrm{d}x &= \Big[x\ln(1+\tan x)\Big]_0^{\frac{1}{4}\pi} - \int_0^{\frac{1}{4}\pi} \ln(1+\tan x)\,\mathrm{d}x \\ &= \frac{\pi}{4}\ln 2 - \frac{\pi}{8}\ln 2 \qquad \text{(using part (iii))} \\ &= \frac{\pi}{8}\ln 2.\end{aligned}$$

So the integral in part (iv) has the same value as the one in part (iii).

Checks

- Part (i): the integrand rises from 0 to 1 and is symmetric about the point $(\frac{1}{4}\pi, \frac{1}{2})$, so its average value is $\frac{1}{2}$ and the integral is half the interval length, $\frac{1}{4}\pi$.
- Part (ii) should be less than half of part (i), because the integrand is below $\frac{1}{2}$ on $[0, \frac{1}{4}\pi]$: indeed $\frac{1}{8}\pi - \frac{1}{4}\ln 2 \approx 0.2194 < 0.3927$.
- Numerical quadrature (Simpson's rule) gives 0.785398, 0.219412, 0.272198 and 0.272198 for the four parts, matching $\frac{1}{4}\pi$, $\frac{1}{8}\pi - \frac{1}{4}\ln 2$ and $\frac{1}{8}\pi\ln 2$ (twice).

Common slips

- Forgetting that the substitution $u = a - x$ swaps the limits; the minus sign from $\mathrm{d}u = -\mathrm{d}x$ swaps them back.
- In part (ii), trying to swap sin and cos as in part (i): on $[0, \frac{1}{4}\pi]$ the reflection is $x \mapsto \frac{1}{4}\pi - x$, not $\frac{1}{2}\pi - x$.
- Writing $\int \tan x\,\mathrm{d}x = \ln\cos x$; it is $-\ln\cos x$ (or $\ln\sec x$).

Beyond STEP

Part (iii) is a classic: the same reflection trick evaluates integrals such as $\int_0^{\frac{1}{2}\pi} \ln \sin x \, \mathrm{d}x = -\frac{1}{2}\pi \ln 2$, which can't be found from an antiderivative in elementary functions. At university these symmetry arguments sit alongside contour integration and differentiation under the integral sign as tools for definite integrals with no elementary antiderivative. For many more examples in the STEP spirit, see Siklos, *Advanced Problems in Mathematics*; for the analysis that justifies substitution and integration by parts, see Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.